EXPERIMENT-10

PIGLATINMODES,PROGRAMS

OBJECTIVE:

a. RunthePig LatinScriptsto findWord Count.
b. RunthePigLatinScriptstofind amaxtempfor eachandeveryyear.

RESOURCES:

VMWare,WebBrowser,4 GBRAM,80 GBHard Disk.

PROGRAMLOGIC:

RunthePig LatinScriptsto findWord Count.

```
lines=LOAD '/user/hadoop/HDFS_File.txt'AS (line:chararray);
words=FOREACHlinesGENERATEFLATTEN(TOKENIZE(line))asword; grouped
= GROUP words BY word;
wordcount=FOREACHgroupedGENERATEgroup,COUNT(words);
DUMP wordcount;
```

RunthePigLatinScriptstofind amaxtempfor eachandeveryyear

```
--max_temp.pig:Findsthemaximumtemperaturebyyear
records = LOAD 'input/ncdc/micro-tab/sample.txt'
AS(year:chararray,temperature:int, quality:int);
filtered_records=FILTERrecordsBYtemperature!=9999AND
(quality==0ORquality==1ORquality ==4ORquality==5ORquality==9); grouped_records
= GROUP filtered_records BY year;
max_temp=FOREACHgrouped_recordsGENERATEgroup,
MAX(filtered_records.temperature);
DUMPmax_temp;
```

INPUT/OUTPUT:

(1950,0,1)

(1950,22,1)

(1950,-11,1)

(1949,111,1)

Scanned with OKEN Scanner

PRE-LABVIVA QUESTIONS:

1. Listoutthehomologous fig?
2. Classify Pig Latin organs and sin fig?

Scanned with OKEN Scanner

LAB ASSIGNMENT:

1. Analyzing average stock prices from the stock data using Apache Pig

POST-LAB VIVA QUESTIONS:

1. Discuss the modes of Pig scripts?
2. Explain the Pig Latin application flow?

Scanned with OKEN Scanner

EXPERIMENT-11

HIVE

OBJECTIVE:

InstallationofHIVE.

RESOURCES:

VMWare, WebBrowser, 1GBRAM, HardDisk80GB.

PROGRAMLOGIC:

InstallMySQL-Server

1) Sudoapt-getinstallmysql-server
2) ConfiguringMySQLUserNameandPassword
3) CreatingUserandgrantingallPrivilege
s Mysql –uroot –proot
Createuser<USER_NAME>identifiedby<PASSWORD>
4) ExtractandConfigureApacheHiv
e tar xvfz apache-hive-
1.0.1.bin.tar.gz
5) MoveApacheHivefromLocaldirectorytoHomedirectory
6) SetCLASSPATHinbashrc
Export HIVE_HOME = /home/apache-hive
ExportPATH=$PATH:$HIVE_HOME/bin
7) Configuringhive-default.xmlbyaddingMySQLServerCredentials

```
<property>
<name>javax.jdo.option.ConnectionURL</name>
<value>
jdbc:mysql://localhost:3306/hive?createDatabaseIfNotExist=tr
ue
</value>
</property>
<property>
<name>javax.jdo.option.ConnectionDriverName</name>
<value>com.mysql.jdbc.Driver</value>
</property>
<property>
<name>javax.jdo.option.ConnectionUserName</name>
<value>hadoop</value>
</property>
<property>
<name>javax.jdo.option.ConnectionPassword</name>
<value>hadoop</value>
```


Scanned with OKEN Scanner

</property>

8) Copyingmysql-java-connector.jartohive/libdirectory.

Scanned with OKEN Scanner

INPUT OUTPUT:

```
hive> create table log_data(i date dt string ( time dt log s sitename st string s
ipaddress st string,i url st string url query st string ip address st string user agent st string
s_status1 int,status2 int,s bytes int c bytes int time taken int);
OK
Time taken: 0.231 seconds
hive> show tables;
OK
log_data
Time taken: 0.074 seconds, Fetched: 1 row(s)
hive> desc log_data;
OK
i_date                  string              None
i_time                  string              None
s_sitename              string              None
s_computer_name         string              None
i_url                   string              None
url_query               string              None
ip_address              string              None
user_agent              string              None
status1                 int                 None
status2                 int                 None
s_bytes                 int                 None
c_bytes                 int                 None
```

PRE-LABVIVA QUESTIONS

1. Define explain the term aggregation and filtration?
2. List out the features of Hive?

LAB ASSIGNMENT:

1. Analyze the Weblog using Apache Hive

POST-LABVIVA QUESTIONS

1. Explain the built-in functions?
2. Discuss the Hive Web Interface Architecture?

Scanned with OKEN Scanner

INPUT/OUTPUT:

```
administrator@ubuntu: ~
d yet. Please use TIMESTAMP instead
hive> create table log_data(l_date string,l_time string,s_sitename string,s_comp
uternamе string,l_uri string,uri_query string,ip_address string,user_agent strin
g,status1 int,status2 int,s_bytes int,c_bytes int,time_taken int);
OK
Time taken: 0.331 seconds
hive> show tables;
OK
log_data
Time taken: 0.074 seconds, Fetched: 1 row(s)
hive> desc log_data;
OK
l_date                  string                  None
l_time                  string                  None
s_sitename              string                  None
s_computername          string                  None
l_uri                   string                  None
uri_query               string                  None
ip_address              string                  None
user_agent              string                  None
status1                 int                     None
status2                 int                     None
s_bytes                 int                     None
c_bytes                 int                     None
```

PRE-LABVIVAQUESTIONS:

1. InHive,explaintheterm„aggregation"anditsuses?
2. ListouttheDatatypesinHive?

LAB ASSIGNMENT:

1.AnalyzetwitterdatausingApacheHive.

POST-LABVIVA QUESTIONS:

1. ExplaintheBuilt-inFunctionsinHive?
2. DescribethevariousHiveDatatypes?

Scanned with OKEN Scanner